



Effects of rate and amount of propionic acid infused into the rumen on feeding behavior of Holstein cows in the postpartum period

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ABSTRACT

The objective of this study was to determine the effects of 2 amounts of propionic acid (PA) infused intraruminally at 2 rates of infusion at the initiation of meals on the feeding behavior of Holstein cows in the postpartum period. We hypothesized that the amount and length of time of infusions would interact to affect feeding behavior: rapid infusion of a higher dose of PA would result in larger meal size with greater time between meals compared with a slower rate, whereas faster infusion of a lower dose of PA would reduce meal size and the time between meals compared with a slower infusion of the same dose. Eight ruminally cannulated, multiparous Holstein cows were used in a 4 × 4 Latin square design experiment. Cows were blocked by parturition date and randomly assigned to treatment sequence within square. Treatments were infusion of 2.5 L of 0.5 M (HI) or 0.2 M (LO) solutions of PA at initiation of meals over 5 min (FST) or 15 min (SLW) for 12 h following feed delivery. Contrary to our hypothesis, no interaction between amount and rate of infusion was detected for any feeding behavior parameter measured. The FST treatments did not affect dry matter intake or metabolizable energy intake compared with SLW. The FST treatments tended to increase meal length compared with SLW (28.1 vs. 22.7) but did not affect meal size. The FST treatments tended to decrease total eating time (108 vs. 122 min/12 h) but did not affect meal frequency compared with SLW. The HI treatments decreased dry matter intake (7.4 vs. 11.5 kg/12 h) and total metabolizable energy intake (22.5 vs. 29.1 Mcal/12 h) compared with LO by decreasing meal frequency (5.8 vs. 7.5 meals/12 h). The HI treatments decreased eating time (103 vs. 127 min/12 h) compared with LO but did not affect meal size. Further

research is warranted on the effects of the temporal supply of propionate on propionate metabolism and feeding behavior.

Key words: feed intake, hepatic oxidation theory, satiety, dairy cow

INTRODUCTION

Propionic acid (PA) from ruminal fermentation plays an important role in controlling feed intake in ruminants fed cereal grains. Research has shown that DMI depression prepartum is greater for cows consuming high-grain diets than for those consuming high-forage diets (Coppock et al., 1972). In the postpartum (PP) period, DMI depression was greater not only for cows fed more starch (Dann and Nelson, 2011) but also for cows fed more fermentable starch sources at the same level of inclusion (Albornoz and Allen, 2018). Propionate is the primary end product of rumen starch digestion and one of the primary fuels affecting satiety during meals (Allen, 2000).

Continuous intraruminal infusion of sodium propionate depressed total metabolizable energy intake (MEI) by cows in the PP period in a dose-dependent manner compared with isomolar solutions of sodium acetate (Oba and Allen, 2003). Continuous abomasal infusion of PA (0.41 mol/h) also decreased total MEI 13% compared with no infusion by decreasing meal frequency (Gualdrón-Duarte and Allen, 2018). When infused at spontaneous meals, PA decreased feed intake by decreasing both meal size and meal frequency for cows in the PP period (Maldini and Allen, 2018) and in mid lactation (Choi and Allen, 1999).

Langhans and Scharrer (1992) and Friedman (1995) demonstrated that hepatic oxidation of fuels signals satiety in rats, and Koch et al. (1998) showed a temporal relationship between feed intake and hepatic energy status, supporting the idea that oxidative metabolism in the liver is involved in feed intake regulation. Allen (2000) presented evidence that the same mechanism likely applies to ruminants. Propionate is an obligate anaplerotic metabolite, stimulating hepatic oxidation of acetyl CoA (Gualdrón-Duarte and Allen, 2017),

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which is a likely mechanism for its hypophagic effects (Allen, 2014).

In a previous study, we infused 1.25 mol of PA over 5 min (**FST**) and 15 min (**SLW**) at spontaneous meals to determine the temporal effects of intraruminal infusion of PA on the feeding behavior of cows in the PP period (Maldini and Allen, 2018). We hypothesized that a greater flux of PA into the rumen and propionate to the liver during meals would stimulate hepatic oxidation and satiety sooner compared with a lower flux. However, FST increased meal size and decreased meal frequency compared with SLW. Given our results, we speculated that faster supply of propionate might have saturated the pathway for propionate metabolism, resulting in a lower first-pass extraction by the liver compared with SLW. Furthermore, the propionate that bypassed the liver likely extended anaplerosis of the TCA cycle and hepatic oxidation of acetyl CoA over time, increasing the intermeal interval and decreasing meal frequency. Because both PA treatments resulted in a substantial and similar reduction of DMI, we thought that infusion of a lower dose of PA over FST or SLW might have different results.

The objective of this experiment was to determine the effects of the temporal supply of PA on feeding behavior of cows in the PP period by evaluating the effects of the amount of PA infused and the length of infusion at meals. We hypothesized that the amount and length of infusions would interact to affect feeding behavior: faster infusion would increase meal size and the time between meals compared with slower infusion for the high dose (1.25 mol), similar to our previous experiment, but with a low dose (0.5 mol), faster infusion compared with slower infusion would decrease meal size and the time between meals. If the faster infusion rate of the high dose, compared with the slower infusion rate, decreased hepatic oxidation by temporally exceeding the metabolic capacity of the liver to extract propionate, we reasoned that with a lower dose, the hepatic capacity of extraction would be limited to a lesser extent and the faster rate of infusion might stimulate hepatic oxidation sooner within meals.

MATERIALS AND METHODS

Animals, Housing, and Diets

Eight multiparous Holstein cows from the Michigan State University Dairy Cattle Teaching and Research Center were ruminally cannulated at least 40 d prepartum and used in this experiment. The Institutional Animal Care and Use Committee at Michigan State University approved all experimental procedures in this

study. Cows were monitored after parturition and randomly assigned to a treatment sequence in a duplicated 4×4 Latin square design experiment, balanced for carryover effects. Four cows with the closest range in days PP were selected for each block of this experiment and adapted to the feeding behavior stalls at least 2 d before the first infusion day. Each cow was housed in an individual tie-stall for the duration of the experiment. Cows were milked twice daily at 0530 and 1730 h, in their stalls. Cows averaged 6.3 ± 2.0 DIM, 690 ± 50 kg BW, and 3.0 ± 0.46 BCS for block 1 and 10.3 ± 5.7 DIM, 680 ± 84 kg BW, and 2.75 ± 0.50 BCS for block 2, on the first day of infusion.

The experimental diet consisted of corn silage, ground corn, alfalfa silage, soybean meal, alfalfa hay, soy hulls, and a vitamin and mineral mix, and it was formulated to provide 25% forage NDF, 23% starch, and 16% CP (Table 1). Diet was fed once a day (1200 h) at 110% of expected intake, from the day of parturition until the end of the experiment. Between 1000 and 1200 h each day, cows were not allowed access to feed soorts and feed offered could be weighed and sampled.

Experimental Treatments

Treatments were 2.5-L solutions at high (0.5 M, **HI**) or low (0.2 M, **LO**) concentration of PA (99.5%, Kemira Industries, Inc., Des Moines, IA) infused into the rumen over 5 min (FST) or 15 min (SLW) at the conditioned meal (first meal after feeding) and at each spontaneous

Table 1. Ingredients and nutrient composition of experimental diet (% of dietary DM, except for DM)

Item	Value
Ingredient	
Corn silage	30.3
Ground corn	18.3
Alfalfa silage	14.9
Soybean meal	13.7
Alfalfa hay	11.9
Soy hulls	7.02
Vitamin and mineral mix ¹	3.93
Nutrient composition	
DM	56.1
OM	91.7
Starch	23.2
NDF	33.4
Forage NDF	25.6
CP	16.1
MP	10.6

¹Vitamin and mineral mix contained 12.8% Ca, 0.1% P, 11.4% NaCl, 10.9% Na, 0.1% Mg, 14 mg/kg Co, 250.0 mg/kg Cu, 9.9 mg/kg I, 745.0 mg/kg Fe, 994.0 mg/kg Mn, 7.5 mg/kg Se, 1,093.0 mg/kg Zn, 148.7 kIU/kg of vitamin A, 24.8 kIU/kg of vitamin D₃, and 737 IU/kg of vitamin E.

meal. Infusions began at feeding, and feeding behavior was recorded for 12 h following feeding. Propionic acid amounts were chosen based on previous experiments (Maldini and Allen, 2018), and infusion rates were selected in an attempt to maximize treatment differences, while constraining the SLW treatment to the expected length of a typical spontaneous meal. Production rates of PA in the rumen for cows consuming normal or low-roughage diets have been reported to range from 13 to 36 mol/d for cows consuming 12 to 15 kg of DM/d (Sutton et al., 2003; Bauman et al., 1971). In a literature review, Allen (1997) reported that the amount of total short-chain fatty acid production ranges from 42 to 115 mol/d for lactating cows. Because propionate concentration in the rumen can increase from 15 to 45% of total fermentation depending on the digestibility of the diet, PA production could be as high as 52 mol/d. Based on our previous data, we expected that cows would have 9 to 11 meal events per day (Maldini and Allen, 2018). Therefore, an infusion of 1.25 and 0.5 mol of PA would be within the expected range of ruminal PA production and unlikely to exceed the physiological amount produced per meal.

Solutions of PA were infused into the rumen using peristaltic pumps (FPU401, Omegaflex Peristaltic Pump, Omega Engineering, Norwalk, CT), programmed to infuse a total of 2.5 L of solution at each meal independent of PA concentration or rate of infusion. The rate of PA infusion into the rumen was calculated to be 0.25 mol/min for FST and 0.084 mol/min for SLW for the HI treatment (1.25 mol per infusion) and 0.1 mol/min for FST and 0.034 mol/min for SLW for the LO treatment (0.5 mol per infusion). We included a rest day between infusion days to reduce carryover effects of treatment. Each cow received 1 treatment combination (HI-FST, HI-SLW, LO-FST, or LO-SLW) on each infusion day, and all cows received all treatments by the end of the experiment. Each block was completed in 10 d: 2 d for adaptation, 4 infusion days, and 4 rest days. All cows within a block were infused on the same days, and all cows completed the experiment by 20 d PP.

Data and Sample Collection and Analysis

Diet ingredients were sampled daily and composited across days of the experiment within each block. Fecal samples were collected every 8 h in a 24-h period (1400, 2200, and 0600 h) on rest days and frozen at -20°C for later analysis. Diet ingredients, Orts, and fecal samples were dried in a 55°C forced-air oven for 72 h and analyzed for DM concentration. Ingredient samples were ground with a Wiley mill (1-mm screen; Arthur H. Thomas, Philadelphia, PA) and analyzed for DM, ash,

NDF, CP, and starch content as previously described by Maldini and Allen (2018).

Digestibility and energy concentration of the diet were determined using methods described by Harvatine and Allen (2006). Digestibility values were used to estimate true digestible energy (DE) of the diet and ME from the diet was calculated according to NRC (2001) as $\text{ME (Mcal/kg)} = 1.01 \times \text{DE (Mcal/kg)} - 0.45$, based on measured DE. Metabolizable energy of the infusion was calculated according to the amount of PA infused, assuming 365.03 kcal/mol of propionate (Lebedeva, 1964). Metabolizable energy intake was calculated as $\text{MEI (Mcal/d)} = \text{ME} \times \text{DMI}$, and total MEI = MEI diet + ME infusion.

Feeding behavior was recorded every 5 s for 12 h during infusion days using a computerized data acquisition system (Dado and Allen, 1993). The system was programmed to trigger infusion pumps at the initiation of each new meal according to Maldini and Allen (2018). Briefly, variation in weight of feed remaining in mangers was calculated, and when the running standard deviation exceeded the threshold of 0.9 kg, an “eating flag” was triggered. When 5 eating flags were triggered within 100 s, infusion of the PA solution would be initiated and last for 5 or 15 min according to treatment. Triggering infusions based on real-time monitoring of feeding behavior resulted in a 12-h average of 9.6 infusions per cow for LO and 7.9 infusions per cow for HI, with 2.3 false positives (infusions without meals) and 0.06 false negatives (meals without infusions) per cow per day with no treatment differences.

Feeding behavior data were analyzed to quantify the number of meals, meal size, meal length, inter-meal interval, eating time, eating rate, hunger ratio, and satiety ratio. We calculated hunger ratio as the weight of meal divided by the premeal interval, and satiety ratio as the weight of meal divided by postmeal interval (Forbes, 2007). Drinking behavior data were analyzed to quantify drink frequency and total water intake. Total liquid intake was calculated by adding the total water intake to the amount of PA infused into the rumen. The moles of PA infused were calculated based on the amount of liquid infused and the concentration of the solution of PA infused in each treatment. We chose to evaluate the effects of PA infusion during the first 12 h after feeding because previous research in our laboratory demonstrated that PA infusions decreased DMI in PP cows only over the first 4 h after feeding and not the remaining 20 h (Stocks and Allen, 2013). Therefore, by evaluating feeding behavior for 12 h we expected to fully capture treatments effects.

Statistical analyses were conducted using JMP Pro 14 software (SAS Institute Inc., Cary, NC), using the mixed model:

Table 2. Effect of intraruminal infusion of 2.5-L solutions of propionic acid (PA) at 0.5 M (HI) or 0.2 M (LO) over 5 (FST) or 15 (SLW) min at the conditioned and spontaneous meals for 12 h after feeding on feeding behavior, DMI, and energy intake for cows in the immediate postpartum period

Item	Infusion				SEM	P-value		
	0.2 M (LO)		0.5 M (HI)			SLW vs. FST	HI vs. LO	Interaction
	FST	SLW	FST	SLW				
DMI, kg/12 h	10.8	12.3	6.48	8.26	1.15	0.13	<0.01	0.88
Meal frequency, meals/12 h	6.75	8.25	5.57	6.00	0.64	0.20	0.03	0.47
Meal length, min/meal	27.7	21.0	28.5	24.3	2.43	0.07	0.46	0.66
Meal size, kg/meal	1.66	1.49	1.42	1.42	0.17	0.67	0.42	0.67
Eating interval, min	73.6	67.2	104	106	10.0	0.85	<0.01	0.71
Eating time, min/12 h	124	130	92.9	114	6.91	0.05	<0.01	0.26
Eating rate, kg of DM/min	0.065	0.072	0.052	0.055	0.004	0.30	<0.01	0.74
Hunger ratio ^{1,2}	0.026	0.029	0.017	0.012	0.018	0.51	<0.01	0.24
Satiety ratio ^{1,2}	0.037	0.047	0.030	0.030	0.21	0.58	0.15	0.61
ME intake, Mcal/12 h								
Diet ³	26.8	28.5	18.7	20.5	3.12	0.46	<0.01	0.99
Infusion ⁴	1.29	1.53	2.94	2.90	0.13	0.42	<0.01	0.24
Total ⁵	28.1	30.1	21.7	23.4	3.11	0.44	0.01	0.97

¹Data analyzed after log-transformation. SEM reported in the analyzed scale.

²Hunger ratio = meal kg of DM/premeal interval; satiety ratio = meal kg of DM/postmeal interval.

³ME intake from the diet was calculated according to NRC (2001). ME (Mcal/kg) = $(1.01 \times \text{DE}) - 0.45$, based on actual digestibility of diets. ME intake (Mcal/d) = ME $\times$ DMI. DE = digestible energy.

⁴ME intake from the infusion was based on energy density of 0.365 Mcal/mol of ME for PA (Lebedeva, 1964).

⁵Total ME intake = ME intake diet + ME infusion.

$$Y_{ijklm} = \mu + C_i(B_j) + P_k(B_j) + B_j + T_l + L_m + T_l L_m + e_{ijklm},$$

where Y_{ijklm} = the response variable; μ = overall mean; $C_i(B_j)$ = random effect of cow i nested within block j ; $P_k(B_j)$ = fixed effect of period k nested within block j ; B_j = fixed effect of block j ; T_l = fixed effect of treatment concentration l (HI, LO); L_m = fixed effect length of infusion m (FST, SLW); $T_l L_m$ = interaction of treatment concentration and length of infusion; and e_{ijklm} = residual. Treatment effects were declared significant at $P < 0.05$, and tendencies for treatment effects were declared at $P < 0.10$.

RESULTS AND DISCUSSION

Contrary to our hypothesis, no interaction of treatment was detected for any of the variables measured (Table 2). The FST treatments did not affect DMI or MEI compared with SLW. The FST treatments tended to increase meal length compared with SLW (28.1 vs. 22.7; $P = 0.07$) but did not affect meal size (1.49 kg/meal, $P = 0.67$). The FST treatments tended to decrease total eating time (109 vs. 122 min/12 h; $P = 0.05$) but did not affect meal frequency ($P = 0.20$) compared with SLW. The HI treatments decreased DMI by 36% (7.4 vs. 11.5 kg/12 h; $P < 0.01$) and total MEI by 23% (22.5 vs. 29.1 Mcal/12 h; $P = 0.01$) compared with LO by decreasing eating time (103 vs. 127 min/12 h; P

< 0.01), increasing time between meals (70.4 vs. 104.9; $P < 0.01$), and decreasing meal frequency (5.8 vs. 7.5 meals/12 h; $P = 0.03$). The HI treatments reduced eating rate (0.053 vs. 0.069 kg/min; $P < 0.01$) and decreased hunger ratio (0.015 vs. 0.027; $P = 0.07$) compared with LO but did not affect meal size ($P = 0.42$). The ME infused each day was greater for HI compared with LO (2.92 vs. 1.41 Mcal, $P < 0.01$) despite the greater number of infusions for LO over 12 h. The additional energy provided by HI failed to compensate for the reduction in MEI of the diet. Satiety ratio was not affected by treatment.

Drinking behavior of cows is reported in Table 3. The HI treatment decreased total liquid intake 26% compared with LO (39.3 vs. 52.9 L; $P = 0.03$), partly because HI had fewer infusions during a 12-h period, and therefore, less liquid was infused into the rumen than LO (15.4 vs. 19.7 L/12 h; $P < 0.01$), but also because HI tended to decrease water intake compared with LO (19.5 vs. 30.5 L/12 h; $P = 0.07$). Water intake is positively correlated with both DMI and the number of meals per day (Dado and Allen, 1994). Therefore, because HI reduced DMI by 36% compared with LO, a reduction in water intake was expected. Although no significant interaction of treatment was detected, SLW tended to have more liquid infused into the rumen than FST (18.2 vs. 16.6 L; $P < 0.10$) because the number of meals was numerically greater for SLW in the LO treatment.

Table 3. Effect of intraruminal infusion of propionic acid (PA) at 0.5 M (HI) or 0.2 M (LO) over 5 (FST) or 15 (SLW) min at the conditioned and spontaneous meals for 12 h after feeding on drinking behavior of cows in the immediate postpartum period

Item	Infusion				SEM	<i>P</i> -value		
	0.2 <i>M</i> (LO)		0.5 <i>M</i> (HI)			SLW vs. FST	HI vs. LO	Interaction
	FST	SLW	FST	SLW				
Water intake, ¹ L/12 h	31.1	29.9	14.2	24.7	0.29	0.31	0.07	0.24
Infusion, L/12 h	17.9	21.4	15.3	15.5	1.95	0.09	<0.01	0.14
Total liquid intake, ¹ L/12 h	51.3	54.4	36.0	42.6	0.13	0.39	0.03	0.67
Drinking frequency, bouts/12 h	5.13	7.00	4.29	4.25	0.95	0.35	0.08	0.33
Infusions, no./12 h	8.75	10.4	7.84	7.88	0.60	0.15	<0.01	0.17
Moles of PA infused/12 h	3.53	4.18	8.07	7.94	0.34	0.42	<0.01	0.24

¹Data analyzed after log-transformation. SEM reported in the analyzed scale.

The results for HI in this experiment differed from our previous experiment in which FST increased both meal size and eating interval compared with SLW (Maldini and Allen, 2018) despite the same dose of PA and length of infusions being used. Cows in the present experiment were earlier in lactation than the previous experiment (average 8.25 ± 4.49 vs. 13.8 ± 2.9 DMI) and might have been more sensitive to the PA treatments if they were in a greater lipolytic state. The similar effects on meal size (1.42 kg) and eating interval (104 and 106 min) for FST and SLW, respectively, within the HI treatment are consistent with both treatments exceeding the capacity of the liver to extract propionate and a more continuous supply of propionate to the liver after the infusion for both infusion rates.

The amount of PA infused for LO was 40% of that infused in the previous experiment to reduce potential saturation of propionate hepatic uptake reactions that might limit propionate extraction from the blood. We did not detect a significant treatment effect of rate of infusion; however, it is possible that the lower dose of PA infused over FST was sufficient to saturate the propionate metabolism pathway, reducing the first-pass uptake by the liver and extending anaplerosis of the TCA cycle over time. If both rates for HI (0.25 mol/min for FST and 0.084 mol/min for SLW) overwhelmed the capacity of the liver to extract propionate from the blood during meals, resulting in similar effects on meal size and eating interval, then the intermediate infusion rate for LO-FST of 0.1 mol/min likely also exceeded the hepatic capacity to extract propionate during the time course of the meal.

The liver is very efficient in extracting propionate from the blood because of the high activity of propionyl CoA synthetase in ruminant hepatocytes (Ricks and Cook, 1981; Demigné et al., 1986). However, the pathway to hepatic uptake of propionate is saturable, and a high flux of PA from infusion could limit the entry of propionate into the TCA cycle. Mesenteric (Baird et

al., 1980) and ruminal (Berthelot et al., 2002) infusions of propionate decreased the efficiency of propionate extraction by the liver in dairy cows and sheep, respectively, and plasma propionate increased linearly with dose when propionate was infused intraruminally from 0 to 1.3 mol/h for 18 h for cows in the PP period (Oba and Allen, 2003). Although these experiments have shown a limitation in the hepatic capacity to extract propionate from the blood, the specific limiting reactions for propionate metabolism in the liver have not been identified for cows in the PP period. Possible limiting reactions include the conversion of propionate to propionyl CoA, which can be limited by enzyme activity (propionyl CoA synthetase) or availability of CoA, and the conversion of methylmalonyl CoA to succinyl CoA (methylmalonyl CoA mutase reaction), which can be limited by enzyme activity or cofactor (vitamin B₁₂) availability. Daugherty et al. (1986) showed a high correlation between liver B₁₂ and endogenous propionate metabolizing activity in lambs, indicating that a limited availability of B₁₂ could contribute to constraining propionate metabolism. In addition, elevated NADH/NAD ratio can inhibit the malate dehydrogenase reaction, which leads to an accumulation of malate and upstream metabolites, likely inhibiting propionate uptake or the release of malate into the blood.

Different responses in feeding behavior are expected depending upon the amount and temporal supply of propionate to the liver (Maldini and Allen, 2018), the capacity of the liver to metabolize propionate and its metabolites (Gualdrón-Duarte and Allen, 2018), and the availability of acetyl CoA (Piantoni et al., 2015). Whereas anaplerosis from propionate can stimulate acetyl CoA oxidation, decreasing meal length and size, limitations for propionate uptake and metabolism can attenuate these effects and extend the supply of propionate over time, increasing the intervals between meals. In addition, propionate is a potent insulin secretagogue in ruminants (Leuvenink et al., 1997). Although effects

of insulin on feed intake in ruminants are inconsistent, increased plasma insulin concentration during meals might possibly contribute to the removal of fuels from the blood more quickly (Oba and Allen, 2000), consequently reducing hepatic acetyl CoA content available for oxidation in the liver (Allen, 2014). A greater reduction of hepatic acetyl CoA concentration following feeding was previously found to be related to greater feed intake (Piantoni et al., 2015), consistent with the hepatic oxidation theory (Allen et al., 2009).

The current and previous results from our laboratory (Bradford and Allen, 2007; Maldini and Allen, 2018) are the first experiments to evaluate the effects of temporal supply of PA within a meal on feeding behavior. Future research relating temporal supply of fuels from diets to the liver and feed intake through their effects on hepatic energy status will provide insights that may accelerate progress in the area of metabolic control of feed intake, leading to better formulation of diets for lactating cows.

CONCLUSIONS

The control of feed intake in dairy cows is complex. The initiation and termination of a meal is more likely to be controlled by a combination of signals rather than a single signal. However, PA clearly plays a major role as the hypophagic fuel from the diet affecting feed intake in ruminants. Although our findings do not support the hypothesis that the amount of PA infused within a meal and the length of infusions interact to affect feeding behavior of dairy cows, the possibility of a reduction in hepatic metabolism of propionate from limited extraction of plasma propionate still remains. Future evaluation of hepatic metabolism and identification of the limiting reactions to propionate uptake and metabolism will further our understanding of the basic mechanisms by which propionate regulates feed intake in dairy cows and might result in novel nutritional and pharmacological approaches to overcome the challenges of feeding cows in the transition period.

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